Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 30)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 1 . 1 Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

0 1 . 2 Explain how the reserve could be designed to provide a range of niches.

[3 marks]

0 1 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

0 1 . 4 Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

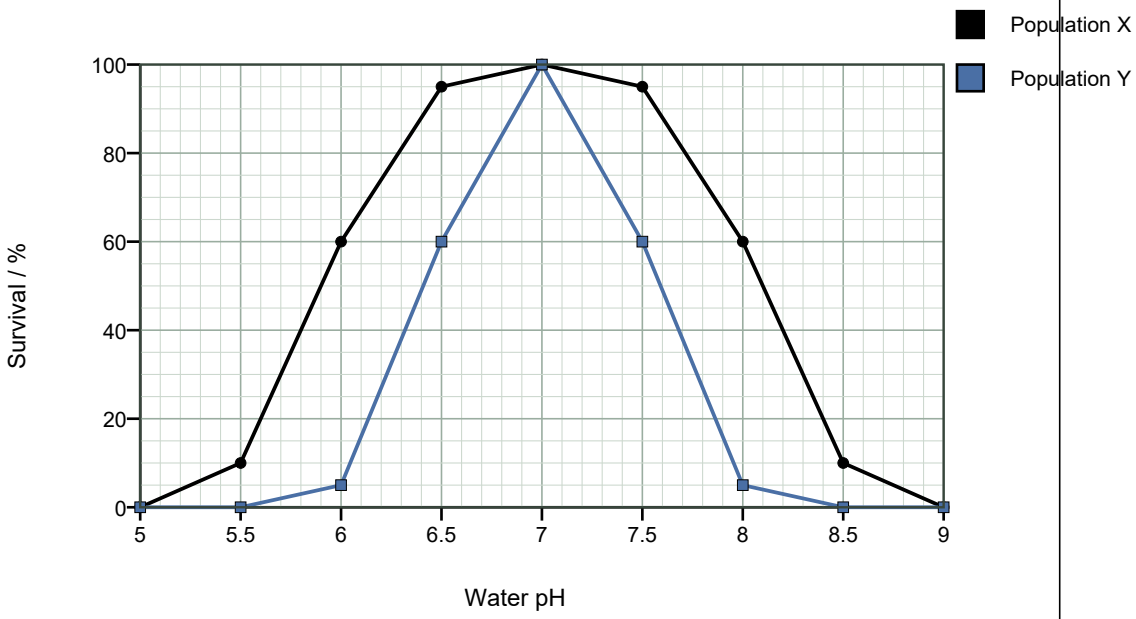
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10

0 2 Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 2 . 1 Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

0 2 . 2 A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

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0 2 . 3 Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

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0 2 . 4 Give **two** biotic factors that affect the distribution of a species.

[2 marks]

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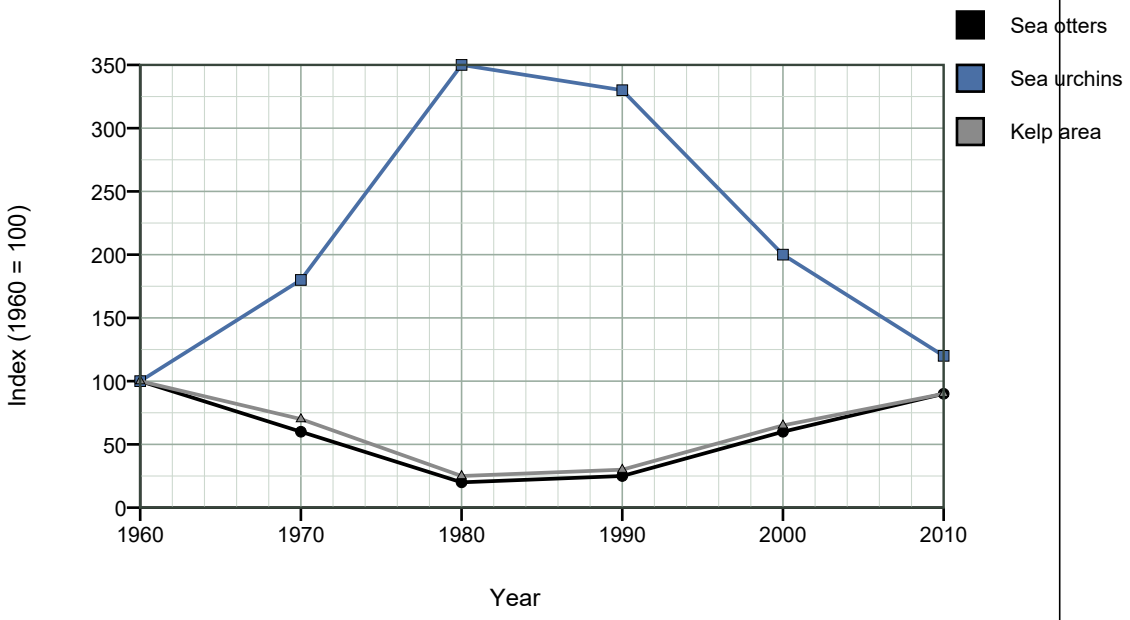
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0 3 Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 3 . 1 Describe the relationship between the three variables shown in Figure 1.

[3 marks]

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0 3 . 2 Explain the relationships shown in Figure 1.

[3 marks]

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0 3 . 3 Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

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0 3 . 4 Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

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0 4 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 4 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

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Answer

0 4 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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5

0 5 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 5 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 5 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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5

0 6 . 1 Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

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0 6 . 2 Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

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0 6 . 3 Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

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Dissolved oxygen .....

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Water flow .....

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Day length .....

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Temperature .....

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0 7 . 1 Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

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0 7 . 2 Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

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0 8 . 1 Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 8 . 2 Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

0 8 . 3 Explain how buying fairtrade food affects producers.

[1 mark]

0 8 . 4 Explain how nitrate from farmland can threaten human health.

[2 marks]

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0 8 . 5 Give **two** agricultural sources of methane.

[2 marks]

1 .....

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2 .....

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0 9 . 1 Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber .....

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Fuel .....

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Food .....

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Fibres .....

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Medicines .....

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0 9 . 2 Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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0 9 . 3 Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

10

1 0 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

1 0 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ____ %

1 0 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

1 0 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

1 0 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

11 . 1 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

OR

11 . 2 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS